

DataSensei Updates 12 — Change Report

Additional Instructor Requirements, Part 1: student and instructor gradebooks, Class Analytics and At-Risk in one place, protected Core Modules, and the three core certificates. Prepared 1 October 2026 against the four-part codebase you uploaded (your working tree, not only the last commit).

What was asked, and what was done

Your PDF starts with “if already implemented, ignore”. Each item was checked against the code first; what was already correct was verified and left alone, and only missing or conflicting parts were changed.

Requirement in the PDF	Delivered
1. Student Gradebook. A student sees only their own grades, results and permitted feedback.	Done. New page Gradebook (student sidebar, /student/gradebook): per class, every published assessment with score, percentage, pass or not passed against the assessment's pass mark, state (graded, late, awaiting grade, held for integrity review, missing, in progress, not open yet), attempts used, the instructor's overall feedback and the number of comments on answers, and a link to the existing result page.
Changing URLs, parameters, IDs or API calls must not reveal another student's grades; privacy enforced in the backend.	Done. The page reads only the signed-in student's id; it accepts no student id at all (student_id, user_id, id are ignored). A class the student is not enrolled in returns 404. <i>Already in place and verified, unchanged:</i> the result page and My Submissions refuse another student's attempt (403). Tested.
2. Instructor Gradebook. Only students enrolled in the instructor's own classes; score, percentage, completion state, feedback/status; no access to other classes.	Done. New page Gradebook (instructor sidebar): a grid of enrolled students × assessments with percentage, score and state, student and class averages; choosing an assessment lists each student's counted attempt, score, percentage, passed or not, state, attempts and the full feedback, with a link to the existing submission page. Another instructor's class or assessment returns 404.
3. Merge Class Analytics and At-Risk into one instructor area.	Done. The sidebar has one item, Class Analytics & At-Risk . The old At-Risk Alerts page is gone; /instructor/risk now opens the same class on the at-risk list (other classes' ids are dropped). The dashboard's at-risk card and counts now come from the same live rules.
Simple bar graph of Low / Moderate / High performance.	Done. “Class performance” on the Overview: three bars (red, amber, green) with the number and share of graded students; students with no graded assessment are reported separately as “Not yet graded”. The Students tab gained a Performance column and filters for each group.
Thresholds from the agreed rules or a configurable setting; no invented hard-coded thresholds.	Done. config/class_analytics.php, set from .env: Low below 70 (the project pass mark), High from 90 (the score the performance clustering already calls “Exceptional Performer”), Moderate in between. Every at-risk number is there too. Invalid values fall back to the defaults.
At-risk students in the same area with reasons/evidence: low assessment scores, low module mastery, repeated failed attempts, configured indicators.	Done, building on the rules Class Analytics already had. Low assessment scores is now the Low performance group (so the graph and the list agree); low module completion, missing assessments and low challenge average were kept; repeated failed attempts now also cover MCQ challenges and assessments, not only coding problems. Each student is listed with every reason that applies and their group.
4. Protect the 24 Core Modules with a database flag, not by title.	Done. modules.module_type (core / custom) and a permanent modules.module_key (core-01-... to core-24-...). The existing 24 modules are marked by the migration; their built-in challenges carry challenges.core_module_key on every version.
Core title and identity cannot be changed; no hard delete (disabled in the UI and rejected by the backend).	Done. The title field is read-only for core modules and the server refuses a change; the Delete button is disabled and destroy() refuses. The same rules sit in the Module model, so no other code path can rename or delete a core module.

Requirement in the PDF	Delivered
Content still updatable through the versioning workflow; activation/deactivation without erasing progress or certificates.	<i>Already in place and verified:</i> the module editor (content) and Publish / Unpublish both still work for core modules; unpublishing keeps every progress row and certificate. Tested.
New modules default to Custom; custom modules editable, deleted only without dependents, otherwise archived.	Done. Every new module is custom (the type cannot be posted). A custom module is deleted only when it has no learner progress, no attempts or coding submissions on its challenges and no class assignments of them; otherwise the page offers Archive (hidden, everything kept) and Restore .
Custom modules never part of the “All 24” certificates.	Done. The certificates list core keys only. Tested with a completed custom module.
5. Certificates: reuse existing progress, issue only after a server-side eligibility check, historically stable; badges stay for small events.	Done. Eligibility is read from module_user, challenge_attempts, challenge_user and coding_submissions (nothing new is recorded). Checked on the server after a lesson completes a module, after a passed MCQ attempt, after a fully solved coding challenge, and when the Certificates page opens. Each certificate stores its requirement version and a snapshot. Achievements (badges) are untouched.
5.1 Core 24 Module Completion.	Done. All 24 Core Modules completed (every lesson).
Core 24 Challenge Completion: 24 designated core challenge requirements; instructor or custom challenges not added.	Done, as you chose: the Newbie-level MCQ challenge of each Core Module, passed with 70% or more (any attempt, any version).
Core Coding Challenge Completion: only the built-in coding challenges under core modules; custom-module and instructor coding challenges excluded; issued certificates preserved if the set changes.	Done, as you chose: the Newbie-level coding challenge of each Core Module that has one (18 on your seeded curriculum), every problem solved. Adding a coding challenge to a Core Module later creates requirement version 2; certificates issued under version 1 stay exactly as issued.
Certificate Builder for instructors, interface requirements, backend rules (scope list).	Not in this delivery: these are described in Part 2 of the requirements, which was not included in the upload.

Decisions you answered, applied exactly

Question	Your answer	Where it is applied
Which challenges count for Core 24 Challenge Completion	Newbie level, one MCQ per Core Module	CertificateService::requirementItems(); pass mark 70% from ChallengePathUnlockService
Which coding challenges count	Newbie level, 18 challenges	Only Core Modules that have one; every problem solved
Low / Moderate / High thresholds	<70 / 70-89 / 90+	config/class_analytics.php (ANALYTICS_LOW_BELOW, ANALYTICS_HIGH_FROM)
Score the groups use	Assessment average	Each student's best graded attempt per assessment; no graded work = “Not yet graded”

Screens


DataSensei
Instructor Workspace

Teaching

Dashboard

Classes

Class Work

Module Library

Challenge Pool

Challenge Builder

Assessments

Question Bank

Table of Specifications

Anti-Cheat

Anti-Cheat Events

Submissions

Analytics

Class Analytics & At-Risk

Gradebook

Reports

Account

Profile

Sign Out

Class

Data Science, DS 4A

Open class

Overview

Students

Modules

Assessments

Challenges

Coding Challenges

Enrolled students

3

Active students

2

Class activity in the last 14 days

Assigned modules

2

Module completion

33.3%

Assessment performance

69.2%

Average best score, 41.7% completed, 4 missing

Challenge performance

33.3%

Completed, average best score 65%

Coding challenge performance

33.3%

Completed, 56.3% of tests passed

Class performance

Low (Below 70%)

1 student, 50%

Moderate (70% to under 90%)

1 student, 50%

High (90% or more)

0 students, 0%

Grouped by each student's assessment average (best graded attempt on each assessment). 1 student has no graded assessment yet and is shown as Not yet graded.

At-risk students

2 of 3

Students who may need attention. A student is listed only when one of these rules is true for their saved work; each reason is shown next to their name:

- Missing 2 or more past-due assessments
- Low performance group: assessment average below 70%
- Challenge average below 70%
- 3 or more failed attempts on an unsolved coding problem
- 3 or more failed attempts on a challenge or assessment not yet passed
- Completed under 25% of the modules assigned 7 days or more ago

Lia Student

Low performance, average 55%

Assessment average 55% (Low performance group); Challenge average 50%; Repeated failures on 1 coding problem; Completed 0 of 2 modules

Tom Student

No graded assessment yet

Missing 3 assessments; Completed 0 of 2 modules

Completion

Modules completed

33.3%

Assessments completed

41.7%

Challenges completed

33.3%

Coding challenges completed

33.3%

Scores

Assessment average

69.2%

Challenge average

65%

Coding tests passed

56.3%

Class Analytics & At-Risk: one sidebar item; the Low / Moderate / High graph and the at-risk students with their reasons side by side, above the existing completion and score bars.

My Gradebook

Your own scores, results and instructor feedback for the homework, quizzes and examinations in your classes. Only you can see this page; your classmates' grades are never shown.

Class

All my classes

Data Science, DS 4A

Instructor: Ana Instructor. Your average: 83.3% over 3 graded assessments.

Assessment	Due	Score	Percentage	Status	Feedback
Loops Worksheet Homework	Sep 28, 2026, 11:45 AM	8 / 10	80% Passed (pass mark 70%)	Graded 1 of 2 attempts used	Good use of loops. Next time add a docstring. View result
Functions Worksheet Homework	Sep 29, 2026, 11:45 AM	8 / 10	80% Passed (pass mark 70%)	Graded 1 of 2 attempts used	— View result
Midterm Quiz	Sep 30, 2026, 11:45 AM	9 / 10	90% Passed (pass mark 70%)	Graded 1 of 2 attempts used	— View result
Upcoming Worksheet Homework	Oct 06, 2026, 11:45 AM	—	—	Not started	—

Student Gradebook: only the signed-in student's own scores, results, attempts and feedback, class by class, with links to their own result pages.

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Page 3

Gradebook

Grades of the students enrolled in your classes: score, percentage, completion state and feedback for every published assessment. Only your own classes are shown.

Class

Data Science, DS 4A

Assessment details

All assessments (grid)

Open

Data Science, DS 4A

3 students, 4 assessments

Each cell shows the counted attempt (the best graded attempt), its percentage and its state. Pass marks use each assessment's passing score, or 70% when none is set. Choose an assessment above to read the feedback given to each student.

Student	Loops Worksheet Homework, due Sep 28, 2026	Functions Worksheet Homework, due Sep 29, 2026	Midterm Quiz Assessment, due Sep 30, 2026	Upcoming Worksheet Homework, due Oct 06, 2026	Average
Lia Student lia student-2umbk@school.test	60% 6 / 10, late	Missing	50% 5 / 10	Not started	55%
Sam Student sam student-nmddi@school.test	80% 8 / 10, feedback given	80% 8 / 10	90% 9 / 10	Not started	83.3%
Tom Student tom student-ucqev@school.test	Missing	Missing	Missing	Not started	—
Class average	70%	80%	70%	—	

Instructor Gradebook: the instructor's own class, every enrolled student against every published assessment, with student and class averages.

Loops Worksheet					Homework, due Sep 28, 2026, 11:46 AM, pass mark 70%		
← Back to the Data Science grid							
Student	Score	Percentage	Result	State	Attempts	Feedback	
Lia Student lia student-qcrkh@school.test	6 / 10	60%	Not passed	Graded (late) Turned in Sep 30, 2026, 11:46 AM	1 of 2	—	Open
Sam Student sam student-jaryc@school.test	8 / 10	80%	Passed	Graded Turned in Sep 27, 2026, 11:46 AM	1 of 2	Good use of loops. Next time add a docstring.	Open
Tom Student tom student-9mho5@school.test	—	—	—	Missing	0 of 2	—	

Instructor Gradebook, one assessment: score, percentage, passed or not, state, attempts and feedback for each student.

Modules

25 modules (24 core, 1 custom) across 4 year levels. Students unlock published modules in the order shown here; drafts are skipped. A new module also gets one locked challenge on each of the 2 challenge levels. Core Modules keep their title and cannot be deleted; new modules are Custom modules.

Add module

Year 1

Order	Module	Content	Status	XP	Type	Module set	Actions
1	Basics of Python Programming Core module 1.	1 section 0 learning outcomes	Published	100	Standard	Core Protected	↓ Preview Edit Unpublish Delete
2	Basics of Statistics Core module 2.	1 section 0 learning outcomes	Published	100	Standard	Core Protected	↑ ↓ Preview Edit Unpublish Delete
3	Introduction to Data Science Core module 3.	1 section 0 learning outcomes	Published	100	Standard	Core Protected	↑ ↓ Preview Edit Unpublish Delete
4	Mathematical Analysis I Core module 4.	1 section 0 learning outcomes	Published	100	Standard	Core Protected	↑ ↓ Preview Edit Unpublish Delete

DataSensei Modules: each module marked Core or Custom; the Delete button of a Core Module is disabled.

Module details

What students see on the Modules page, where DataSensei Modules are grouped by year level and unlocked in order.

Title

Basics of Python Programming

Year level

Year 1

Core Module: the title and identity cannot be changed. Its content can still be edited.

Description

Core module 1.

XP reward

100

Module type

Standard module

Coding exercises

☐ This module has coding exercises

Changing this later does not create or remove challenges.

Published

Cancel

Preview Module

Save Changes

Unpublish

Core Module

This is one of the 24 Core Modules (core-01-basics-of-python-programming). Its title and identity are fixed and it cannot be deleted; the core certificates depend on it. You can still edit its content here, and publish or unpublish it: unpublishing hides it from learners and keeps their progress and certificates.

Delete module

Editing a Core Module: the title is read-only, the content stays editable, publish and unpublish still work, and deletion is disabled with the reason.

Certificates

Three certificates mark the big milestones of the DataSensei core curriculum. They are checked on the server from the progress you have saved and issued as soon as you qualify. Badges for smaller wins stay under Achievements.

Core 24 Module Completion

Completed all 24 DataSensei Core Modules, every lesson of each module.

9 of 24 done

Every lesson of each of the 24 Core Modules completed.

► Show each requirement

Core 24 Challenge Completion

Passed the Newbie-level MCQ challenge of each of the 24 Core Modules with 70% or more.

14 of 24 done

The Newbie-level MCQ challenge of each of the 24 Core Modules passed with 70% or more (any attempt, any version).

► Show each requirement

Core Coding Challenge Completion

Solved every problem of the built-in Newbie-level coding challenge of each Core Module that has one.

3 of 18 done

Every problem solved in the Newbie-level coding challenge of each Core Module that has one (all problems of one version).

► Show each requirement

Certificates (student sidebar): the three core certificates with the learner's progress, checked on the server.

← Certificates

DATASENSEI CERTIFICATE

Core 24 Module Completion

This certifies that

Lia Student

Completed all 24 DataSensei Core Modules, every lesson of each module.

Issued
October 1, 2026

Certificate number
DS-MOD-20261001-WTTQOGMF

Requirements
Version 1, 24 items

Print or save as PDF

An issued certificate, printable or saved as PDF from the browser. It shows the snapshot taken when it was issued.

Welcome back, Ana Instructor.

Review live class, coursework, and at-risk data from the classes assigned to you.

Create class

Create assessment

Open analytics

Active classes

1

owned by you

Unique learners

3

across active classes

Published work

4

published assessments

Average score

69%

assessment average of graded work

Class modules

2

assigned to your classes

At risk

2

unique learners flagged by the at-risk rules

Class performance

Detailed analytics →

Class	Learners	Avg. score	Low / Moderate / High	At risk
Data Science DS 4A	3	69%	1 / 1 / 0 1 not yet graded	2

Analytics

Recently updated work

Manage work →

Loops Worksheet

Homework, Data Science, 0 seconds ago

Published

At-risk learners

Class Analytics & At-Risk →

Lia Student

Data Science, Low performance. Assessment average 55% (Low performance group); Challenge average 50%; Repeated failures on 1 coding problem; Completed 0 of 2 modules

At risk

Tom Student

Data Science, not yet graded. Missing 3 assessments; Completed 0 of 2 modules

At risk

Instructor Dashboard: average score, performance groups and at-risk learners now come from the same live rules as Class Analytics.

How it fits the existing system

One source for grades. Class Analytics, both gradebooks and the dashboard read the same records. A score counts as a grade only when it is final: graded, or late and fully scored. An attempt held for an integrity review, or still waiting for an essay grade, is shown with that state but no longer counts toward averages or groups (App\Support\Reports\SubmissionOutcome). Before this change, a held attempt (stored score 0) pulled a student's average down.

Configurable thresholds. App\Support\Reports\PerformanceBands reads config/class_analytics.php and checks every value; nothing about groups or at-risk numbers is fixed in a page. The rule list shown above the at-risk students is generated from the same values.

At-risk merge. ClassProgress already computed live “needs attention” rules for Class Analytics; the separate At-Risk Alerts page used stored performance snapshots with different rules. Those two are now one list. The “Recalculate” action is kept for the snapshots other pages still read, and returns to the merged page.

Core identity. App\Support\CoreCurriculum lists the 24 agreed keys. The titles in it are used once, to find the existing core rows (with a fallback for a core module renamed before this update), and never again. A new version of a core challenge inherits the core key through the shared content code; an instructor's challenge never does. A core challenge's content code, level and type cannot change, and its last version cannot be deleted, so the certificates never point at something missing.

Certificates. Three system definitions (certificate_definitions), their versioned requirement lists (certificate_requirement_sets) and issued certificates (user_certificates, one per learner and certificate, with requirement version and snapshot). A requirement version is frozen once a certificate has been issued under it; a later change becomes the next version. Certificates are issued only to learners, never twice, and a notification is sent. A certificate can only be opened by its holder (404 otherwise).

Database (MySQL 5.5)

One additive migration, 2026_10_02_000001_add_core_modules_and_certificates, guarded with information_schema checks like the existing ones: plain columns with defaults or NULL, longText instead of JSON, short indexed strings (at most 100 characters, within MySQL 5.5's index limit with utf8mb4), no foreign keys, no generated columns. It adds modules.module_type, modules.module_key (unique), modules.archived_at, challenges.core_module_key, and the tables certificate_definitions, certificate_requirement_sets and user_certificates, then marks the core rows and defines the three certificates. Running it again is a no-op.

Run on a copy of your fully seeded curriculum: 24 modules marked core with their keys, all 210 built-in challenge rows (MCQ and coding, five levels) given their core key, requirement lists of 24, 24 and 18 items; a second run reported nothing to migrate, and the sync command then marked nothing new.

After unzipping

Step	Command or action
1. Unzip over C:\xampp\htdocs\ so the DataSensei folder merges.	—
2. Apply the migrations (mark the Core Modules, define the certificates, repair the module progress table).	<code>php artisan migrate</code>
3. Reload configuration.	<code>php artisan config:clear</code>
4. Optional: issue the certificates learners have already earned from earlier work. (Otherwise each learner gets them the next time they make progress or open Certificates.)	<code>php artisan datasensei:core-sync --award</code>
5. Optional: delete the old At-Risk view, which nothing uses any more.	<code>resources\views\instructor\risk\index.blade.php</code>
6. Optional: change thresholds in .env (keys listed in .env.example), then config:clear.	<code>ANALYTICS_LOW_BELOW=70 ANALYTICS_HIGH_FROM=90</code>
If you re-run the curriculum or challenge seeders later	They mark the core rows themselves; for a single module seeder run <code>php artisan datasensei:core-sync</code>

Verification

Check	Result
PHPUnit, full suite	1,620 passed, 12,244 assertions, 34 skipped (the Docker-only sandbox tests, as before), 0 failed
New tests for this delivery	29 tests in 5 files (Updates12GradebookPrivacyTest, Updates12ClassAnalyticsAtRiskTest, Updates12CoreModulesTest, Updates12CertificatesTest, Updates12LegacyProgressTableTest): an older module progress table, IDOR attempts on the gradebooks and result pages, other instructors' classes, configured thresholds and fallbacks, the merged page and redirect, core title/identity/delete guards in the controller and the model, archive and restore, core challenge versions, all three certificates end to end (including through the lesson-complete route), version stability, holder-only access, learners only
Existing tests	All still pass, including the Class Analytics, reports, dashboard, module manager and “no pills / no schema queries in page code” rules
Migration on the seeded curriculum (SQLite copy)	Applied; 24 core modules, 210 core challenge rows, 3 certificates (24 / 24 / 18 items); second run is a no-op
MySQL DDL	Generated with Laravel's MySQL grammar and checked by hand for MySQL 5.5: no JSON, no FKs, indexed strings $\leq$ 100 characters
Every Blade template compiles	Passed (<code>php artisan view:cache</code>)
Screens	Rendered by the application itself for each role and photographed in Chromium (the screens above)

Not verified here

A MySQL 5.5 server itself (none could be installed in this environment; the SQL above is what Laravel sends to MySQL), XAMPP on Windows, and a click-through in a live browser session. The screens were rendered from the application's real responses for real users and photographed, not mocked.

Fix after delivery: “Unknown column opened_at” on the instructor dashboard

On your MySQL database the table `module_library_progress` already existed before DataSensei Updates 8, so the Updates 8 migration (which only creates it when it is missing) left it with its old columns, without `opened_at`, `last_opened_at`, `completed_at` and `class_id`. Class Analytics already read those columns; this update made the instructor dashboard read them too, which is where the error appeared. The same gap also affected the instructor Reports and the class module pages.

Two changes fix it. A new migration, `2026_10_02_000002_repair_reporting_table_columns`, adds every missing column to `module_library_progress` (and to `audit_logs`, made the same way) and fills the new columns from the old rows: `opened_at` from `created_at`, `last_opened_at` from `updated_at`, `completed_at` only where the old table marks a completion (`is_completed`, `completed` or `status`). A table that already has the columns is left exactly as it is. Separately, the report queries now read a missing optional column as empty instead of failing (`App\Support\Reports\Columns`), so the pages stay up even before the migration runs. Covered by `Updates12LegacyProgressTableTest`, which rebuilds the old table shape and reproduces the error without the fix.

Files in the ZIP

Only files changed or added in this delivery; folder structure preserved. Unzip over `C:\xampp\htdocs\` so the DataSensei folder merges. One file is no longer used (marked “deleted”; a ZIP cannot remove it, see step 5).

Path	Change
<code>.env.example</code>	modified
<code>app/Console/Commands/SyncCoreCurriculum.php</code>	added
<code>app/Http/Controllers/AdminPublicModuleController.php</code>	modified
<code>app/Http/Controllers/ChallengesController.php</code>	modified
<code>app/Http/Controllers/CodingQuizController.php</code>	modified
<code>app/Http/Controllers/InstructorAnalyticsController.php</code>	modified
<code>app/Http/Controllers/InstructorAtRiskController.php</code>	modified
<code>app/Http/Controllers/InstructorController.php</code>	modified
<code>app/Http/Controllers/InstructorGradebookController.php</code>	added
<code>app/Http/Controllers/LessonController.php</code>	modified
<code>app/Http/Controllers/StudentCertificateController.php</code>	added
<code>app/Http/Controllers/StudentGradebookController.php</code>	added
<code>app/Models/CertificateDefinition.php</code>	added
<code>app/Models/Challenge.php</code>	modified
<code>app/Models/Module.php</code>	modified
<code>app/Models/UserCertificate.php</code>	added
<code>app/Services/CertificateService.php</code>	added
<code>app/Services/Reports/ClassProgress.php</code>	modified
<code>app/Services/Reports/GradebookService.php</code>	added
<code>app/Services/Reports/InstructorReports.php</code>	modified
<code>app/Support/CoreCurriculum.php</code>	added
<code>app/Support/Reports/Columns.php</code>	added

Path	Change
app/Support/Reports/PerformanceBands.php	added
app/Support/Reports/SubmissionOutcome.php	added
config/class_analytics.php	added
database/migrations/2026_10_02_000001_add_core_modules_and_certificates.php	added
database/migrations/2026_10_02_000002_repair_reporting_table_columns.php	added
database/seeder/ClassroomSeeder.php	modified
database/seeder/ModuleChallengesSeeders.php	modified
database/seeder/ModuleCodingChallengesSeeders.php	modified
database/seeder/ModuleSeeder.php	modified
docs/Updates_12_Change_Report.pdf	added
resources/views/admin/modules/_form.blade.php	modified
resources/views/admin/modules/edit.blade.php	modified
resources/views/admin/modules/index.blade.php	modified
resources/views/instructor/analytics/index.blade.php	modified
resources/views/instructor/analytics/student.blade.php	modified
resources/views/instructor/dashboard.blade.php	modified
resources/views/instructor/gradebook/index.blade.php	added
resources/views/instructor/risk/index.blade.php	deleted (remove it)
resources/views/partials/instructor-sidebar.blade.php	modified
resources/views/partials/sidebar.blade.php	modified
resources/views/reports/partials/bars.blade.php	modified
resources/views/student/certificates/index.blade.php	added
resources/views/student/certificates/show.blade.php	added
resources/views/student/gradebook/index.blade.php	added
routes/web.php	modified
tests/Feature/Regression/Updates12CertificatesTest.php	added
tests/Feature/Regression/Updates12ClassAnalyticsAtRiskTest.php	added
tests/Feature/Regression/Updates12CoreModulesTest.php	added
tests/Feature/Regression/Updates12GradebookPrivacyTest.php	added
tests/Feature/Regression/Updates12LegacyProgressTableTest.php	added
docs/Updates_12_Change_Report.pdf	added (this document)